Mohammed Huzaifa

IoT & Embedded Systems Engineer

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SUMMARY

IoT engineer comfortable on both sides of the wire: embedded firmware and sensor hardware on one end, full-stack dashboards and protocol simulation on the other. Prefers building a whole prototype rather than one layer of it.

SKILLS

Embedded & Hardware	Arduino, ESP32, GPIO, PWM, Interrupts, Timers, I2C, SPI, UART, PCB basics, power & battery management
Wireless & Protocols	Wi-Fi, Bluetooth/BLE, LoRa, LoRaWAN, NB-IoT, MQTT, CoAP, TCP/IP
Languages & Tools	Python, C/C++, Java, Git/GitHub, PlatformIO, Edge Impulse, AWS IoT, Azure IoT Hub

PROJECTS

PULSE: Wearable Mood-Responsive Environment Control

ESP32 · MAX30102 · MPU6050 · GSR · I2C · ADC · MQTT · WebSocket

- Built an ESP32 wearable fusing a GSR sensor (skin conductance), a MAX30102 (heart rate/SpO2), and an MPU6050 (6-axis IMU) to infer the wearer's arousal state live. Learned how motion data is what separates a genuine mood spike from someone just moving their hand around.
- Designed a four-stage pipeline (signal smoothing, 60-second personal calibration, weighted Z-score arousal index, hysteresis state machine) and learned the accuracy-vs-stability tradeoff firsthand: raw thresholding hit 86% accuracy but threw 65 false transitions a minute, while adding hysteresis traded down to 79% for zero spurious flips.
- Carried the system through to live environmental response (lighting, audio, climate) and wrote up every equation in an IEEE-format paper. Learned to scope a wellness prototype honestly: no output is framed as a clinical reading or diagnosis.

NetSim: Browser-Based LoRaWAN Network Simulator

LoRaWAN · EU868 duty cycle · ADR (SF7-SF12) · Semtech time-on-air · Confirmed/unconfirmed uplinks

- Modeled 50 battery-powered LoRaWAN devices against 3 gateways with EU868 duty-cycle enforcement, Adaptive Data Rate (SF7-SF12), and Semtech's time-on-air formulas. Learned how spreading-factor choice trades range for airtime and battery life in a real LPWAN deployment.
- Implemented collision modeling with a capture effect and per-radio-state (sleep/RX/TX) power draw, learning how contention and duty-cycle caps actually throttle throughput on constrained wireless networks.
- Built the physics engine to run standalone, with no live hardware or RF transmission required, to make LoRaWAN's normally invisible constraints visible and testable.

Smart Home Hub: Automation Rules Engine with 3D Simulation

Trigger/condition/action rules · Device state model · Per-device tokens · Schema validation · Replay protection

- Built a trigger/condition/action rules engine and learned to wire real device state (signal strength, battery, staleness) directly into rule evaluation, so an automation can't fire a device that's offline or reporting stale data.
- Implemented a command-security layer (per-device tokens, schema validation) and built a working replay-attack demo. Learned concretely how a naive security layer fails, and what schema validation alone doesn't catch.
- Treated security and connectivity as constraints baked into the core logic rather than cosmetic add-ons, learning how that mirrors the way production automation systems actually have to behave.

MINOR & FOUNDATIONAL WORKS

- **Temperature & Humidity Monitor** (2023): a DHT sensor and a display on an Arduino, the first build and the one that taught the rest.
- **Ocean Pollution Detection Sensor** (2024): a waterproof sensor array lowered into water, returning pH and chemical readings live instead of weeks after a lab visit.
- **Smart India Hackathon: Smart Attendance Register** (2024): face-recognition attendance using nothing but a webcam; placed top 20 of roughly 150 competing teams.
- **PIR & GSM Security Alarm** (2024): a passive-infrared sensor and a SIM module that texts a phone when something moves with no WiFi, app, or account needed.

SPECIALIZED TRAINING

Samsung Innovation Campus, IoT Track, Presidency University, Bangalore

Selected through a competitive entrance test (27 of 350 applicants). Coursework: Fundamentals of IoT & Embedded Circuit Systems; Version Control & Collaborative IoT Software Development.

CERTIFICATIONS

- BCG Data Science Job Simulation, BCG X (September 2026)
- Mastercard Cybersecurity Virtual Experience, Mastercard (September 2026)